

Agent-swarm physics: hypothesis compendium

Living status document, regenerated 2026-10-03 17:32 by `infra/summaries/build_summaries.py`. Not the paper. One page per hypothesis follows; click a row to jump to it.

Completion: estimated progress of the research direction (idea 5%, round 1 done 35–50%, holdout run 55–65%, robustness and causal designs 70–85%, settled 90–100%). 5% 25% 50% 75% 100%

Faithfulness (0–5, scoped): how well the model holds for what it claims (3 = descriptive or primary holdout passed; 4 = supported; 5 = faithful and mechanistic). **Usefulness** (0–5, unscoped): what an operator or alignment researcher can do with it, true or not (1 = vocabulary only; 2 = a diagnostic; 3 = a validated monitor or design rule; 4 = a steering lever; 5 = a transferable control knob). 0 1 2 2.5 3 4 5

Ratings are estimates calibrated across hypotheses by the coordinator; rubric in `writeup/hypothesis-pages/RUBRIC.md`.

ID	Hypothesis	Complete	Faithful	Useful	p.
H01	Emergent superagents exist (draft) running. Exploratory round 1 of D3.1.a and D3.2 done (2026-10-03, non-holdout only): rooms are reliably more ordered ...	—	—	—	3
H02	Inferred couplings reflect real influence (draft) #45 confirmation FAILED (locked holdout, run 2026-10-03): the Fine-Tuned Leader ranks 4th of 18 by net outgoing influ...	—	—	—	4
H03	Swarm activity is self-exciting, and its criticality is set by the ... (draft) running. Exploratory round 1 is complete (2026-10-03, non-holdout data only).	—	—	—	5
H04	External forcing reshapes the response kernel, reversibly (draft) confirmatory run on the locked holdout (2026-10-03): C1 FALSIFIED, in the opposite direction. The branching ratio n i...	—	—	—	6
H05	Cutting the cross-room channel lowers entropy production, and rooms... (draft) confirmatory run on the locked holdout (2026-10-03): INCONCLUSIVE by the pre- registered rule. The primary pair DiD at...	—	—	—	7
H07	The RPG forks diverge from a common ancestor at a measurable rate (draft) running. Results are organized by goal period (see "Results by goal period"). Exploratory round 1 (non-holdout: #35 o...	—	—	—	8
H08	Context is the coupling (draft) idea, parked. Opened 2026-10-03 on a misread: the request was about the thermodynamics framing, which is H09. Kept be...	—	—	—	9
H09	Agent swarms have an effective thermodynamics (draft) exploratory rounds 1 (E1–E5) and 2 (E6–E8) done, light. Observable tables built (idle_runs, consolidation_inflow). No...	—	—	—	10
H10	Goals are Legendre pushes: a swarm's response to an assigned goal i... (draft) running (exploratory round 1). Predictions written 2026-10-03 before any real-data run (below).	—	—	—	11
H11	Division of labor vs herding is the sign of a Potts coupling, set b... (draft) exploratory round 1 done (2026-10-03). The sign-by-goal-mode prediction is not supported. $\beta J_{CW} > 0$ (herding) in 11 o...	—	—	—	12
H12	Groupthink is dimensional collapse: few collective modes, shrinking... (draft) running. Exploratory round 1 done 2026-10-03: HH77 partially supported (one Curie–Weiss-like market mode, partly join...	—	—	—	13
H13	Model families carry their own fields, and couple by family (draft) exploratory round 1 done (2026-10-04).	—	—	—	14
H14	Entropy production of behavior-state sequences: per-family arrows o... (draft) specified (promoted 2026-10-03 by Vivian from shortlist 2). Exploratory round 1 running: observables, nulls and predi...	—	—	—	15

ID	Hypothesis	Complete	Faithful	Useful	p.
H15	Semantic information through natural scrambles: which information k... (draft) exploratory round 1 done (2026-10-03, non-holdout only). The pre-registered "memory is load-bearing" predictions fail...	—	—	—	16
H16	Agents get stuck in metastable traps and escape by Kramers kinetics... (draft) specified (promoted 2026-10-03 by Vivian from shortlist 2). Exploratory round 1: synthetic validation done; predictio...	—	—	—	17
H17	Behavior is a Markov state model with a few metastable sets; its mi... (draft) specified (promoted 2026-10-03 by Vivian from HH97). Exploratory round 1 running: ob-servables, nulls and predictions ...	—	—	—	18
H18	Attention dilutes as 1/k: response to a message falls with the numb... (draft) specified (promoted 2026-10-03 by Vivian from HH99). Exploratory round 1: observables, nulls and predictions written ...	—	—	—	19
H19	One curve for all periods: per-period loop gains collapse onto a fu... (draft) running. Exploratory round 1 done (2026-10-04): the pre-registered collapse P1 FAILED, with no common curve: equal-ti...	—	—	—	20
H20	Aging: day-to-day content autocorrelation depends on time since kic... (draft) running (exploratory round 1). Observables, null and predictions written 2026-10-03, before any real-data run (below).	—	—	—	21
H21	The debate week (#12) is a two-sublattice antiferromagnet (draft) running (exploratory round 1, 2026-10-03). Promoted 2026-10-03 by Vivian from HH103.	—	—	—	22
H22	Private, conflicting goals make #51 a spin glass (draft) exploratory round 1. Observables, nulls and predictions written 2026-10-04 00:15 UTC (2026-10-03 PT), before any real...	—	—	—	23
H23	Distilling the village into a fine-tuned leader copies its vocabula... (draft) exploratory round 1 running (2026-10-03). Pipeline reconstructed; observables, null and prediction written below befo...	—	—	—	24
H24	In the forecast week (#21), switching from independent drafts to co... (draft) running (exploratory round 1). Switch-on time determined from the record and predictions written 2026-10-03, before a...	—	—	—	25

H01 Emergent superagents exist

not yet rated

level: hypothesis · pipeline checks 85% · updated —

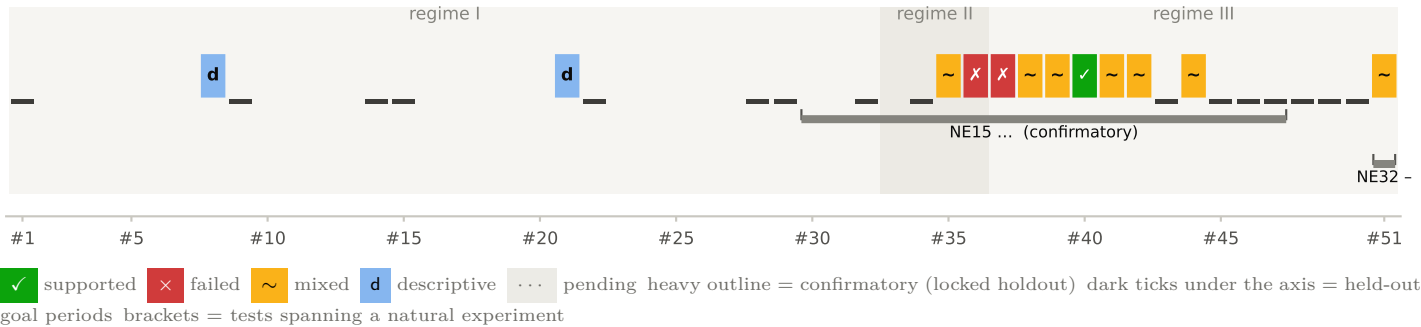
running. Exploratory round 1 of D3.1.a and D3.2 done (2026-10-03, non-holdout only): rooms are reliably more ordered than random groups but miss the pre-registered size; a small, instrument-sensitive coupling residual plus a strong merge effect; the field-d...

WHAT HAPPENED
Not yet summarized. The question: Do groups of agents form emergent superagents: coarse-grained units that act more like one agent than their members do? The main lenses: - a shared ideology as an ordered, low-semantic-entropy state the unit maintains; - Kolchinsky–Wolpert semantic information for which parts of it are load-bearing.

PHYSICAL MODEL
From: physics-models/04-semantic-information. Ideology order parameter: models 11 or 10. Groupings: model 10. Dynamics: models 02 and 09.

Pending.

GOAL PERIODS TESTED 1 supported, 2 failed, 7 mixed, 2 descriptive, 1 pending, 1 n/a



OBSERVABLES
Figure pending.

WHAT THE FIGURE SHOWS
Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS
Pending.
Bottom line. *Pending.*

H02 Inferred couplings reflect real influence

not yet rated

level: hypothesis · pipeline checks 100% · updated —

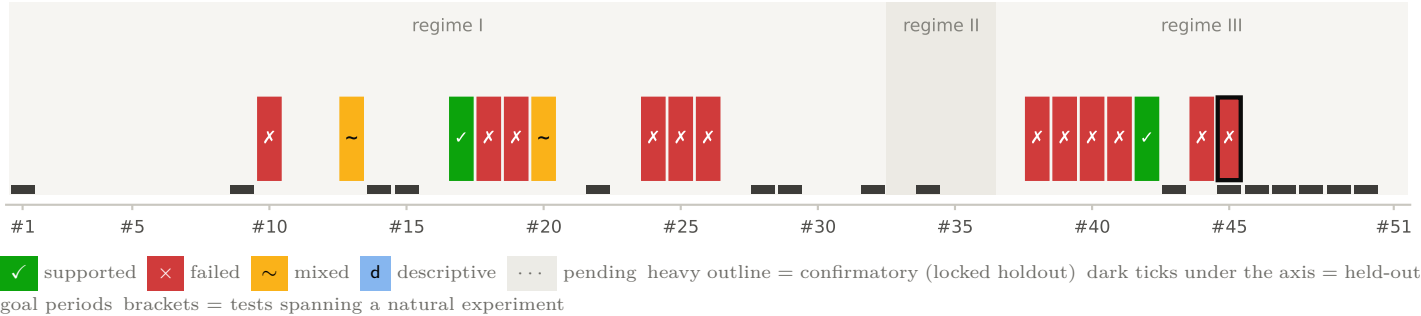
#45 confirmation FAILED (locked holdout, run 2026-10-03): the Fine-Tuned Leader ranks 4th of 18 by net outgoing influence ($z = 0.76$; the rule needed rank 1 and $z \geq 2$). Before that, exploratory round 1 found:

WHAT HAPPENED
Not yet summarized. The question: When we fit Ising couplings to the village’s binned activity, do they measure who influences whom? Or are they artifacts of the schedule, the scheduler, shared model family and finite data? Concretely: - (a) Can the pipeline recover planted couplings and a planted leader from data sampled like the village? - (b) Do individual-objective (mode I) weeks couple less than shared-objective (mode C) weeks, once the strongest nulls are applied? - (c) Does the known leader (#45, held out) come out as the top influencer?

PHYSICAL MODEL
From: physics-models/02-nonequilibrium-ising/ (primary) and physics-models/01-inverse-ising/ (rival / equal-time view). - Spin. $s_i(t) = +1$ if agent i is *active* in 1-min bin t (activity_bins.state ≥ 3 : act or talk), else -1. - KL-1, kinetic Ising (primary). Parallel update, lag one bin: $P(s_i(t+1) = +1 | s(t)) = \sigma(2 H_i)$, with $H_i = h_i + \delta_i(b(t+1)) + J_{ii} s_i(t) + \sum_{\{j \neq i\}} J_{ij} s_j(t)$. - J_{ij} is the influence of j

Pending.

GOAL PERIODS TESTED 2 supported, 12 failed, 2 mixed; holdout: G45 failed



OBSERVABLES
Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS
Pending.

Bottom line. Pending.

WHAT THE FIGURE SHOWS
Pending.

H03 Swarm activity is self-exciting, and its criticality is set by the goal’s coupling mode

not yet rated level: hypothesis · pipeline checks 71% · updated —

running. Exploratory round 1 is complete (2026-10-03, non-holdout data only).

WHAT HAPPENED

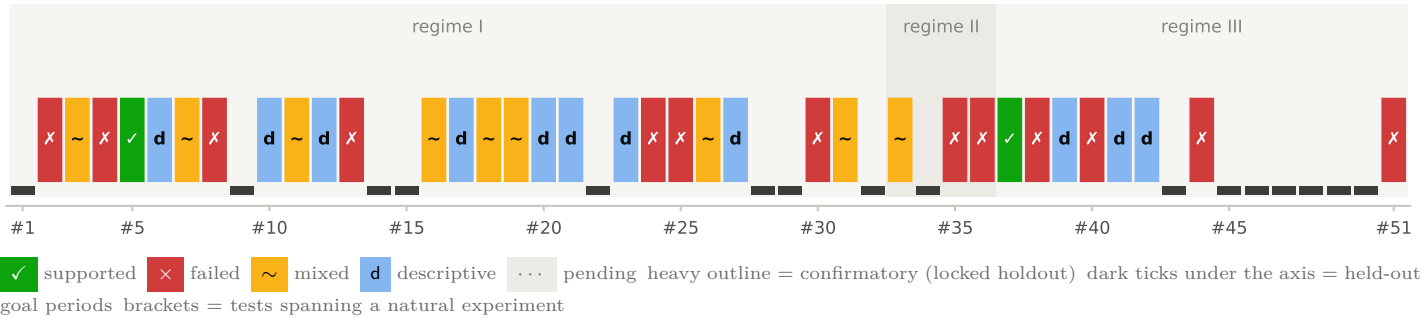
Not yet summarized. The question: How much of the swarm’s activity is self-generated (triggered by earlier agent events) rather than driven by the schedule, goal kickoffs and humans? Is the branching ratio n set by the goal’s coupling mode (shared objective vs. free/holiday, etc.), and does the long private-role era (#51) drift toward criticality ($n \rightarrow 1$) as the roster grows?

PHYSICAL MODEL

From: physics-models/09-hawkes/ (primary), physics-models/02-nonequilibrium-ising/ (directed-influence reading of the cross-agent kernel). Each realization is one village day d , with $t \in [0, T_d]$ in seconds since `win_start`. Excitation never carries across days. M1 (primary, univariate pooled): $\lambda(t) = \mu_B(t) + \sum_{\{x < t\}} \sum_r \gamma_r \delta_r e^{-\delta_r(t-x)} + \sum_{\{t_j < t\}} \alpha \beta e^{-\beta(t-t_j)}$, $n = \alpha$. - Baseline μ_B (the Filimonov–Sornet

Pending.

GOAL PERIODS TESTED 2 supported, 13 failed, 9 mixed, 11 descriptive



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

H04 External forcing reshapes the response kernel, reversibly

not yet rated

level: hypothesis · pipeline checks 100% · updated —

confirmatory run on the locked holdout (2026-10-03): C1 FALSIFIED, in the opposite direction. The branching ratio n is *higher* on 8 h days at all three hours switches; the ABAB contrast is at the edge of week-to-week noise. C2 is inconclusive, C3 untestabl...

WHAT HAPPENED

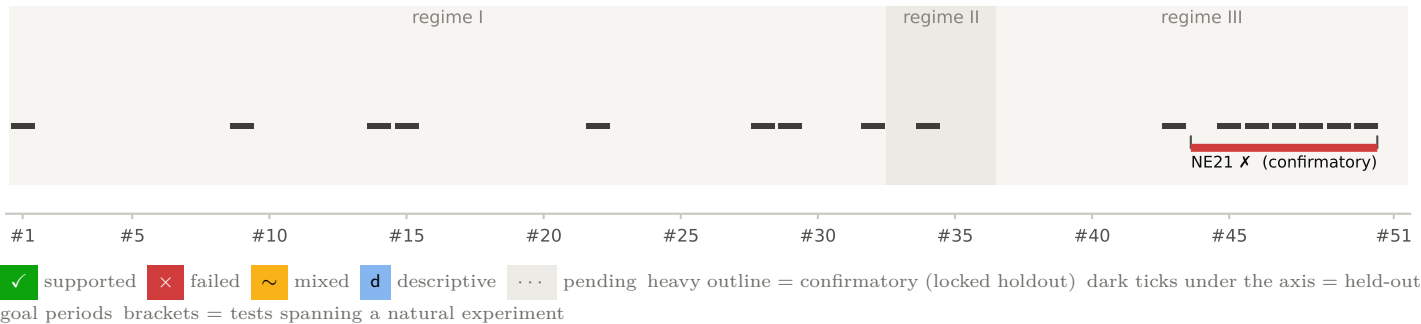
Not yet summarized. The question: Do outside kicks (nudger messages, human messages, goal kickoffs) act on the swarm through a well-defined linear response kernel $G(\tau)$? If they do, does changing the forcing regime (hours $4 \rightarrow 8 \rightarrow 4 \rightarrow 8$ h, NE21; nudger off/on, NE23) re-shape that kernel and the swarm’s branching ratio reversibly, and how far is the swarm from equilibrium by the fluctuation–dissipation (FD) yardstick?

PHYSICAL MODEL

From: physics-models/02-nonequilibrium-ising/ (response function, susceptibility, FD ratio and effective temperature) and physics-models/09-hawkes/ (exogenous and endogenous kernels, branching ratio n). Two views of the same propagator: 1. Minute-binned response (model 02 view). Agent i ’s activity $n_i(m) \in \{0, 1\}$ per minute; a kick at minute m_0 is a field pulse. $G(\tau) = E[n_i(m_0+\tau) \mid \text{kick}] - E[n_i(m_0+\tau) \mid \text{matched no-k}]$

Pending.

GOAL PERIODS TESTED 1 failed; holdout: NE21 failed



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

H05 Cutting the cross-room channel lowers entropy production, and rooms become coupled blocks

not yet rated level: hypothesis · pipeline checks 100% · updated —

confirmatory run on the locked holdout (2026-10-03): INCONCLUSIVE by the pre-registered rule. The primary pair DiD at the 03-16 split (C1) passed: separated pairs lost talk coupling relative to pairs that stayed together. The MF J_out criterion passed. The ...

WHAT HAPPENED

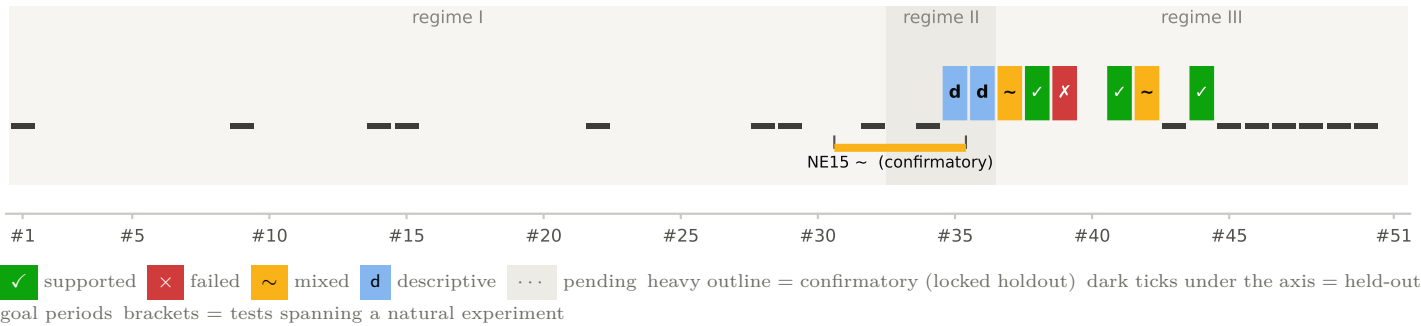
Not yet summarized. The question: When the chat channel between two groups of agents is cut (they are put in different rooms and see only their own room), does the coupling between the groups vanish while coupling inside each group holds, and does the swarm’s statistical irreversibility (entropy production) fall? Equivalently: do rooms behave as coupled blocks, i.e. candidate superagents?

PHYSICAL MODEL

From: physics-models/02-nonequilibrium-ising/ (primary), with 01-inverse-ising (symmetric couplings) and 09-hawkes (rival/complement, not fitted in round 1). - Agents are binary spins on 1-min bins, $s_i(t) = +1$ if the agent acted or talked in the bin ($activity_bins.state \geq 3$), else -1. Secondary spin: talk only ($state = 4$). - Kinetic Ising with parallel updates, $P(s_i(t+1) | s(t)) \propto \exp(s_i(t+1) H_i(t))$, $H_i = h_i(t)$

Pending.

GOAL PERIODS TESTED 3 supported, 1 failed, 3 mixed, 2 descriptive; holdout: NE15 mixed



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

H07 The RPG forks diverge from a common ancestor at a measurable rate

not yet rated

level: hypothesis · pipeline checks 83% · updated —

running. Results are organized by goal period (see "Results by goal period"). Exploratory round 1 (non-holdout: #35 onward) done 2026-10-03: divergence is gradual and measurable at file and function level, punctuated for content (names, numbers), the two fo...

WHAT HAPPENED

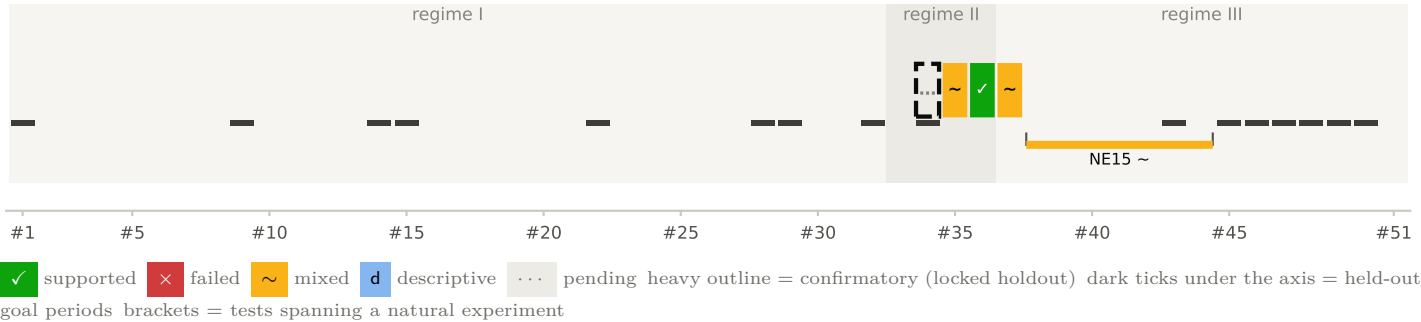
Not yet summarized. The question: On 2026-03-16 the village was split into #best (GPT-5.4, Claude Opus 4.6, Gemini 3.1 Pro) and #rest (everyone else), and each room took its own copy of the RPG that all twelve agents had built during #34. Two isolated subpopulations then evolved two copies of one artifact. How fast, and in which features, do the copies diverge from the common ancestor and from each other? Is the divergence gradual (an inheritance curve with a measurable mutation rate), and does anything cross between the rooms?

PHYSICAL MODEL

From: physics-models/08-copying-vs-transformation/ (primary); 10-potts (rooms as domains) and 09-hawkes (per-fork commit kernels) as secondary (not fitted in round 1). - Channel. A keyed feature (a file path, a numeric parameter at a code path, a named game entity) has value x in the ancestor and value y in a descendant snapshot. The ensemble of keys defines p(x, y). Two channels: vertical (ancestor → lineage k at ti

Pending.

GOAL PERIODS TESTED 1 supported, 3 mixed, 1 pending



OBSERVABLES

Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

WHAT THE FIGURE SHOWS

Pending.

H08 Context is the coupling

not yet rated

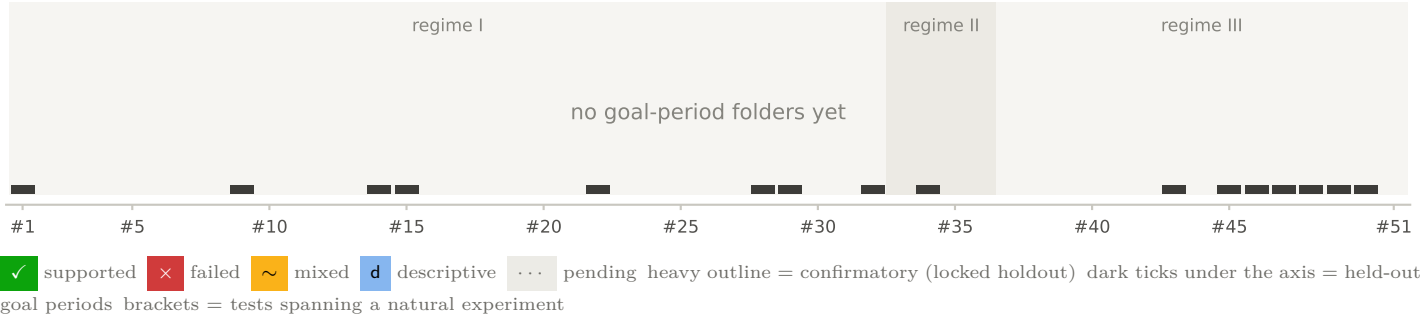
level: hypothesis · pipeline checks 14% · updated —

idea, parked. Opened 2026-10-03 on a misread: the request was about the thermodynamics framing, which is H09. Kept because exposure calibration (C1) matters for every direction, and H numbers are permanent. Nothing has been run.

WHAT HAPPENED

Not yet summarized. The question: Agents interact *only* through what enters their context. So the coupling between agents is, mechanically, *context inclusion*: what each agent saw, and when. How faithfully can we reconstruct that from the logs, and how much of each agent’s behavior does it explain? Every other direction needs a good answer. Contagion, couplings (S1, S4), field vs. coupling (S5) and H01’s natural scrambles all rest on an exposure model.

GOAL PERIODS TESTED none yet



OBSERVABLES

Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

PHYSICAL MODEL

From: none primary; this is a measurement hypothesis that calibrates the exposure input of models 01, 02, 03, 09, 11 and H01.

Pending.

WHAT THE FIGURE SHOWS

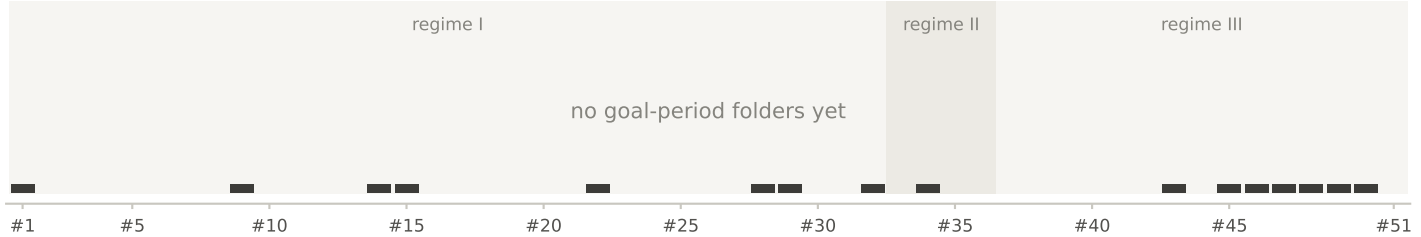
Pending.

H09 Agent swarms have an effective thermodynamics

not yet rated level: hypothesis · pipeline checks 14% · updated —
exploratory rounds 1 (E1–E5) and 2 (E6–E8) done, light. Observable tables built (idle_runs, consolidation_inflow). Not promoted. Opened 2026-10-03 at Vivian's request.

WHAT HAPPENED
Not yet summarized. The question: Can the swarm be described with the working parts of thermodynamics: - free-energy landscapes over states; - temperature-like and field-like control parameters, with susceptibilities; - chemical potentials; - metastable traps and activation barriers; - probability currents and entropy production; - work done by operators, and an equation of state for memory? And do those descriptions predict anything that simpler statistics don't?

GOAL PERIODS TESTED none yet



✓ supported
✗ failed
~ mixed
d descriptive
... pending
heavy outline = confirmatory (locked holdout)
dark ticks under the axis = held-out goal periods
brackets = tests spanning a natural experiment

OBSERVABLES
Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS
Pending.

Bottom line. Pending.

PHYSICAL MODEL
From: models 01, 02 (entropy production, response), 09 (kinetics), 10 (occupancy over categorical states), 06/07 (built-in free energies). The framing is model-agnostic; each T needs its own model choice when promoted.
Pending.

WHAT THE FIGURE SHOWS
Pending.

H10 Goals are Legendre pushes: a swarm's response to an assigned goal is predictable from its unforced fluctuations

not yet rated

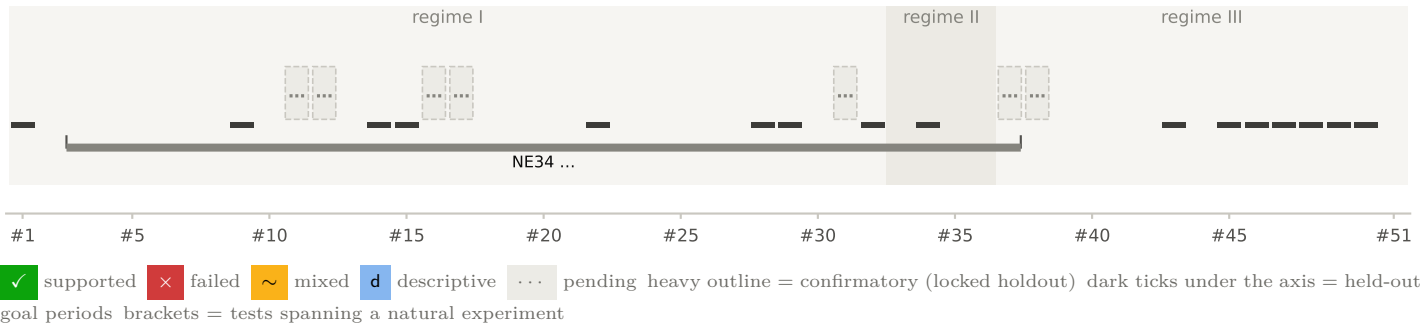
level: hypothesis · pipeline checks 57% · updated —

running (exploratory round 1). Predictions written 2026-10-03 before any real-data run (below).

WHAT HAPPENED
Not yet summarized. The question: Estimate the free energy $F(m)$ of the swarm's alignment m from its fluctuations in field-free (pick-your-own-goal) weeks. Can it predict how the swarm lines up behind an assigned goal, by tilting $G = F - h \cdot m$, without refitting? Practical payoff: forecasting how steerable a swarm is from how it behaves when nobody steers it.

PHYSICAL MODEL
From: physics-models/11-vector-spins (mean-field $O(n)$: $m = L_n(\beta(J_0|m| + h))$ m ; susceptibility $\chi = N \cdot \text{var}(m)$), with physics-models/01-inverse-ising for the scalar Curie-Weiss analogue. One field parameter h per assigned week is fitted to the mean shift; the variance and higher cumulants under the field are then **predicted** (an unfitted test, axis D).
Pending.

GOAL PERIODS TESTED 8 pending



OBSERVABLES
Figure pending.

WHAT THE FIGURE SHOWS
Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS
Pending.
Bottom line. *Pending.*

H11 Division of labor vs herding is the sign of a Potts coupling, set by the goal

not yet rated

level: hypothesis · pipeline checks 76% · updated —

exploratory round 1 done (2026-10-03). The sign-by-goal-mode prediction is not supported. $\beta J_{CW} > 0$ (herding) in 11 of the 14 tested weeks, whatever the mode, including all five weeks predicted to be AF. The exceptions are the three weeks where agents built...

WHAT HAPPENED

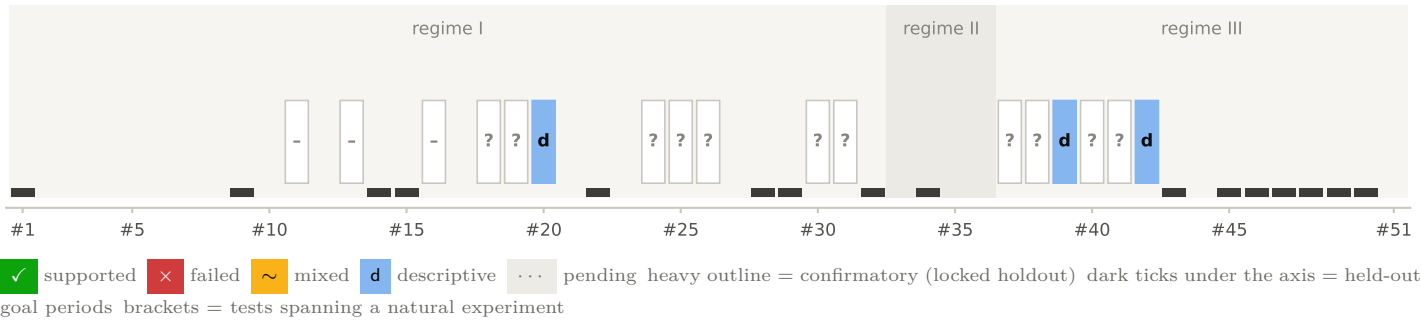
Not yet summarized. The question: When agents work on a shared goal, do they spread across subtasks (antiferromagnetic Potts coupling: division of labor, graph coloring) or pile onto the same project (ferromagnetic: herding)? Does consensus arrive as a first-order jump (mean-field Potts, $q \geq 3$)? Practical payoff: detecting duplicated work and group-think in multi-agent systems.

PHYSICAL MODEL

From: physics-models/10-potts, mean-field forward first. One uniform-coupling q-state Potts model per period, fitted from the dominant option’s share and its fluctuations. It predicts the sign of J by goal mode, and the jump size and hysteresis at consensus events. Full plmDCA couplings are only a secondary check (H02 showed pairwise inference is weak here). H11 variant (Curie–Weiss Potts with a neutral state, in roo

Pending.

GOAL PERIODS TESTED 3 descriptive, 3 n/a



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

H12 Groupthink is dimensional collapse: few collective modes, shrinking participation ratio at consensus

not yet rated

level: hypothesis · pipeline checks 86% · updated —

running. Exploratory round 1 done 2026-10-03: HH77 partially supported (one Curie–Weiss-like market mode, partly joint lulls); HH58 refuted in direction (kickoffs expand dimensionality; self-repetition, not consensus, collapses it). Confirmatory script writ...

WHAT HAPPENED

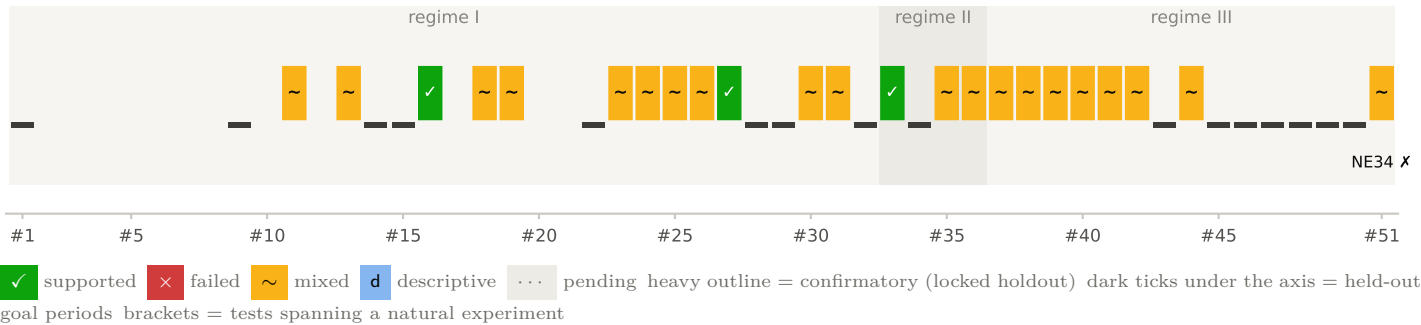
Not yet summarized. The question: How many collective modes does the swarm have beyond random-matrix noise, and does the effective dimensionality of what agents say collapse after kickoffs and during consensus, then re-expand in free weeks? Practical payoff: a cheap, model-free monitor of whether a swarm is 'thinking alike'.

PHYSICAL MODEL

From: physics-models/01-inverse-ising and physics-models/11-vector-spins with random-matrix theory: Marchenko–Pastur edges for $N \times T$ correlation matrices (activity or talk spins per minute; content time series per window), and the participation ratio $PR = (\sum \lambda)^2 / \sum \lambda^2$ of the swarm's whitened embedding covariance per window. The top mode is compared with H02's Curie–Weiss collective co-activation. - Model 01 reading. For e

Pending.

GOAL PERIODS TESTED 3 supported, 1 failed, 20 mixed



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

H13 Model families carry their own fields, and couple by family

WHAT HAPPENED

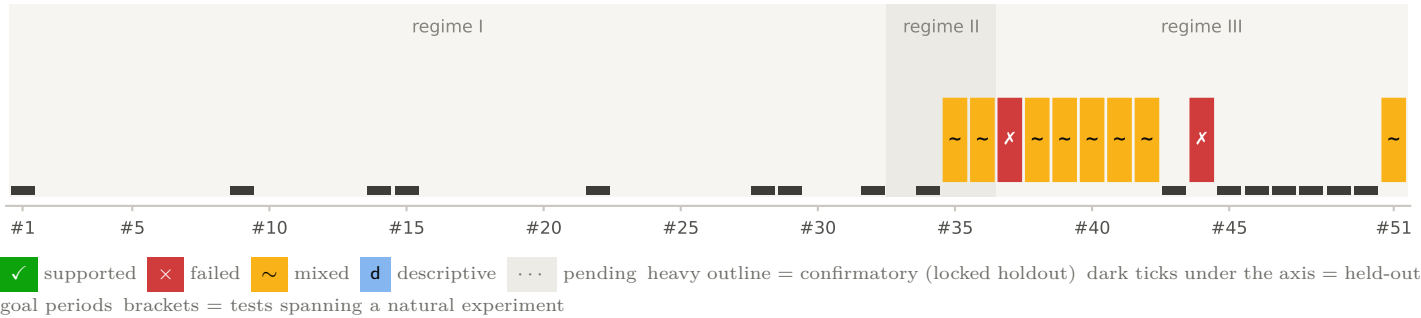
Not yet summarized. The question: Does each lab’s models share a direction in content space (a family field h_f , e.g. the ‘genuinely’ magnetization), beyond the goal field? Is coupling homophilous by family: a $K \times K$ family coupling matrix with $J_{\text{within}} > J_{\text{across}}$? Does family identity persist across room boundaries, or does room identity dominate (H01 D9.2)? Practical payoff: vendor bias and homophily in mixed-model swarms, with about 9 parameters instead of $N \times N$.

PHYSICAL MODEL

From: physics-models/11-vector-spins (fields), physics-models/10-potts, and physics-models/01-inverse-ising / 02-nonequilibrium-ising in K-population mean-field form (family blocks, like H05’s two-block mean field). Lexical magnetizations, such as per-family rates of marker words, come from derived counts only. H13 variant (vector spins with family fields and family-block couplings). For agent i of family $f(i)$, on d_a

Pending.

GOAL PERIODS TESTED 2 failed, 8 mixed



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

H14 Entropy production of behavior-state sequences: per-family arrows of time, and collective irreversibility

not yet rated

level: hypothesis · pipeline checks 57% · updated —

specified (promoted 2026-10-03 by Vivian from shortlist 2). Exploratory round 1 running: observables, nulls and predictions written 2026-10-03 before any real-data run (below).

WHAT HAPPENED

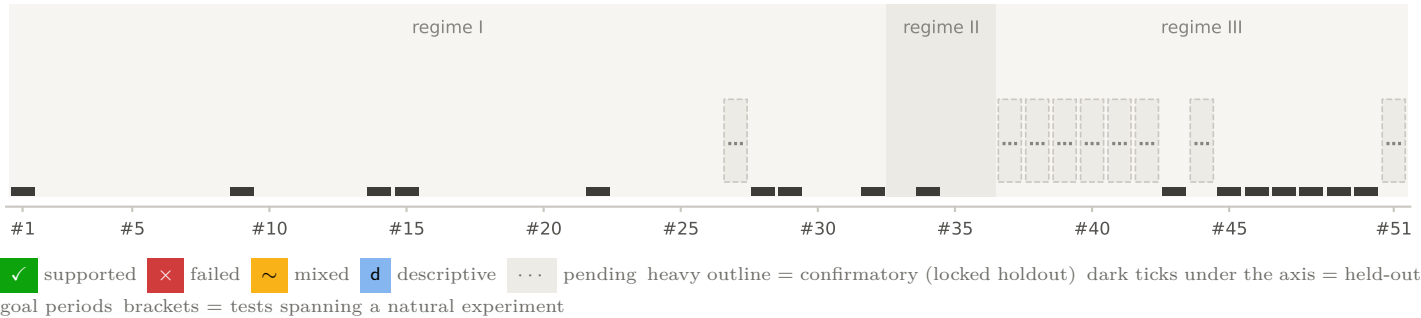
Not yet summarized. The question: 1-minute activity spins were too coarse for entropy production (H05). With richer categorical states (action classes now; Jev behavior states later), how irreversible is each agent’s behavior? Does each model family have its own arrow of time (HH19)? Is the swarm’s joint irreversibility more than the sum over agents (HH67)? Where do the probability currents run (HH56)? Practical payoff: irreversibility as a measure of goal-directed, driven work versus aimless activity.

PHYSICAL MODEL

From: physics-models/02-nonequilibrium-ising generalized to kinetic Potts states (physics-models/10-potts), with the Aguilera–Ito–Kolchinsky MaxEnt estimator and plug-in Markov EP with finite-sample bias correction. H05’s validated analysis/ep.py is imported, not copied (cross-fitted Newton-step bound ep_gauss_crossfit, held-out exact dual ep_heldout). - Single agent: $x_t \in \{1...q\}$, a stationary Markov chain within a g

Pending.

GOAL PERIODS TESTED 9 pending



OBSERVABLES

Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

WHAT THE FIGURE SHOWS

Pending.

H15 Semantic information through natural scrambles: which information keeps agents and the swarm viable

not yet rated

level: hypothesis · pipeline checks 85% · updated —

exploratory round 1 done (2026-10-03, non-holdout only). The pre-registered "memory is load-bearing" predictions failed: no detectable day-scale viability cost of losing up to ~80% of memory (P1) or of starting with an empty memory (P4); the rewrite control...

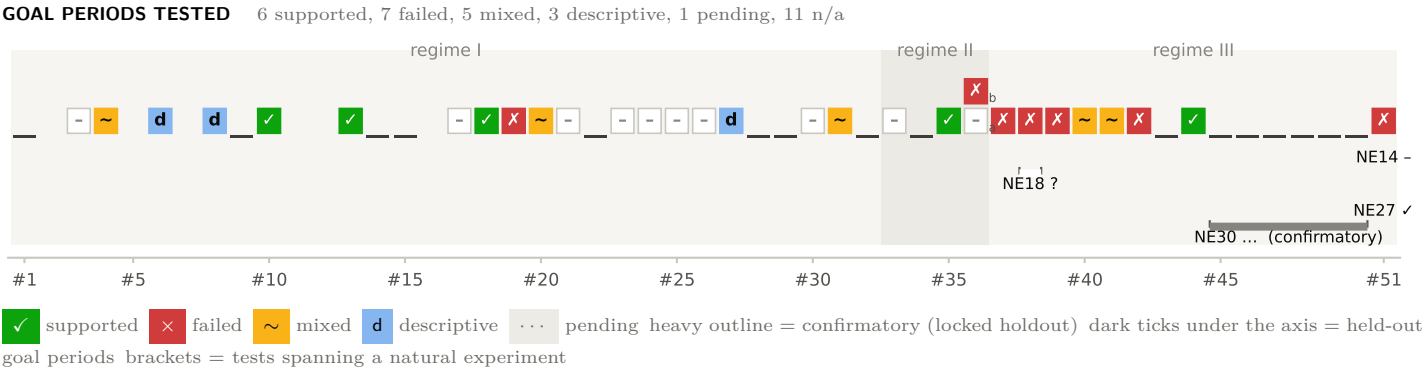
WHAT HAPPENED

Not yet summarized. The question: Kolchinsky–Wolpert: information is semantic, or load-bearing, if scrambling it lowers the system’s viability. We can’t run new swarms or replay agents, but the logs contain natural scrambles: memory loss and resets, a change in consolidation rate (NE14), newcomers with empty memories, room cuts that remove chat exposure, and widened history search (NE18). Which removals lower viability, by how much, and what fraction of the available information is load-bearing (η)? Practical payoff: which context actually keeps a swarm working.

PHYSICAL MODEL

From: physics-models/04-semantic-information (ΔV under scrambling, η = load-bearing fraction), with H01’s architecture.md (viability candidates, D2.6 homeostasis rule for choosing V) and notes/kolchinsky-qa-transcript.md. ### Mapping onto Kolchinsky–Wolpert (KW) - System X: one agent. Its state has three parts with different lifetimes: - long-term memory M (agent_memories; summarized without text in memory_stats), ca

Pending.



OBSERVABLES

Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

WHAT THE FIGURE SHOWS

Pending.

H16 Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier

not yet rated

level: hypothesis · pipeline checks 57% · updated —

specified (promoted 2026-10-03 by Vivian from shortlist 2). Exploratory round 1: synthetic validation done; predictions written below (2026-10-03) before any real-data run.

WHAT HAPPENED

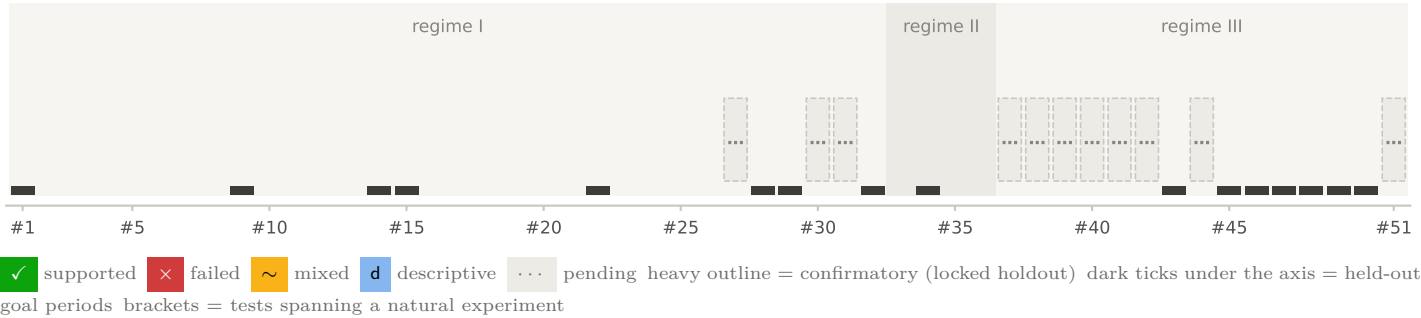
Not yet summarized. The question: Idle loops, repeated-error debugging loops and theory spirals look like local free-energy minima. Do dwell times in these trap states follow Kramers escape, with the rate set by a barrier from Boltzmann inversion of occupancies? Do kicks (nudges, @-mentions, human messages) lower the barrier, with escape hazard rising exponentially in kick strength? At swarm level, does mean-field theory ($\beta J_0 < 1$, as measured) correctly predict *no* collective bistability, so that traps are individual? Practical payoff: predicting when agents get stuck and what gets them out.

PHYSICAL MODEL

From: physics-models/02-nonequilibrium-ising and 01-inverse-ising (mean-field double well $f(m) = -J_0 m^2/2 - h m - T \cdot s(m)$; bistable only when $\beta J_0 > 1$), with Kramers escape along a reaction coordinate, and 11-vector-spins for content-level spirals (not run in round 1). H16 variant (single agent). Each agent's activity is an overdamped coordinate in a double well, $U(x)$ with an idle well and an active well, at effective te

Pending.

GOAL PERIODS TESTED 11 pending



OBSERVABLES

Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

WHAT THE FIGURE SHOWS

Pending.

H17 Behavior is a Markov state model with a few metastable sets; its mixing time is a per-period order parameter

not yet rated

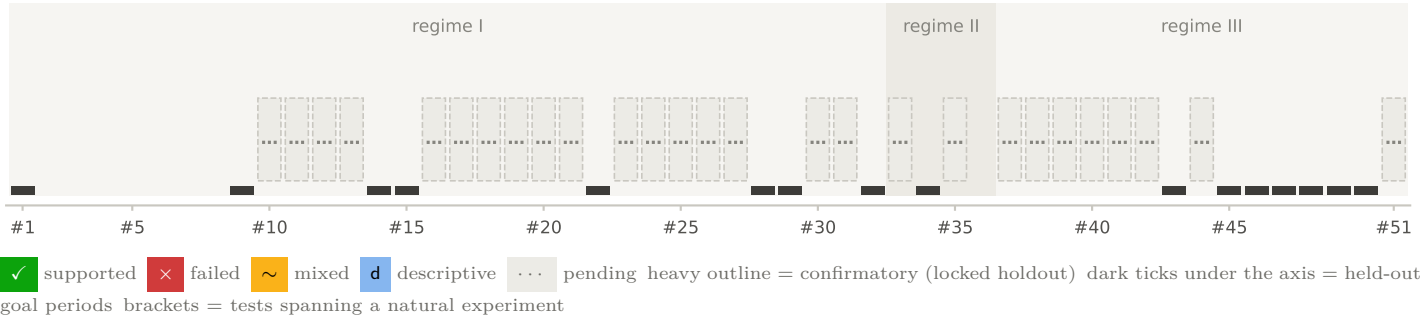
level: hypothesis · pipeline checks 57% · updated —

specified (promoted 2026-10-03 by Vivian from HH97). Exploratory round 1 running: observables, nulls and predictions written 2026-10-03 before any real-data run (below).

WHAT HAPPENED
Not yet summarized. The question: Coarse-grain each agent’s behavior-state sequence into a Markov state model. Do a few slow, metastable sets (‘attractors’ such as coding, debugging loops, existential talk) dominate, as identified by spectral clustering of the transition matrix (PCCA+)? Is the spectral gap, or slowest implied timescale, a per-goal-period order parameter: slowest in stuck periods, fastest in productive ones? Practical payoff: a compact map of where agents get stuck and how fast they move between modes.

PHYSICAL MODEL
From: physics-models/02-nonequilibrium-ising and 10-potts (kinetic Potts as a Markov chain), with Markov-state-model machinery: implied timescales and the Chapman–Kolmogorov test for Markovianity at lag τ , PCCA+ metastable sets, committors between sets. Nonequilibrium: the transition matrix need not satisfy detailed balance, so cycle currents are expected (link to H14).
Pending.

GOAL PERIODS TESTED 27 pending



OBSERVABLES
Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS
Pending.

Bottom line. *Pending.*

WHAT THE FIGURE SHOWS
Pending.

H18 Attention dilutes as 1/k: response to a message falls with the number of messages waiting at the agent's turn

not yet rated

level: hypothesis · pipeline checks 57% · updated —

specified (promoted 2026-10-03 by Vivian from HH99). Exploratory round 1: observables, nulls and predictions written 2026-10-03, before any real-data run.

WHAT HAPPENED

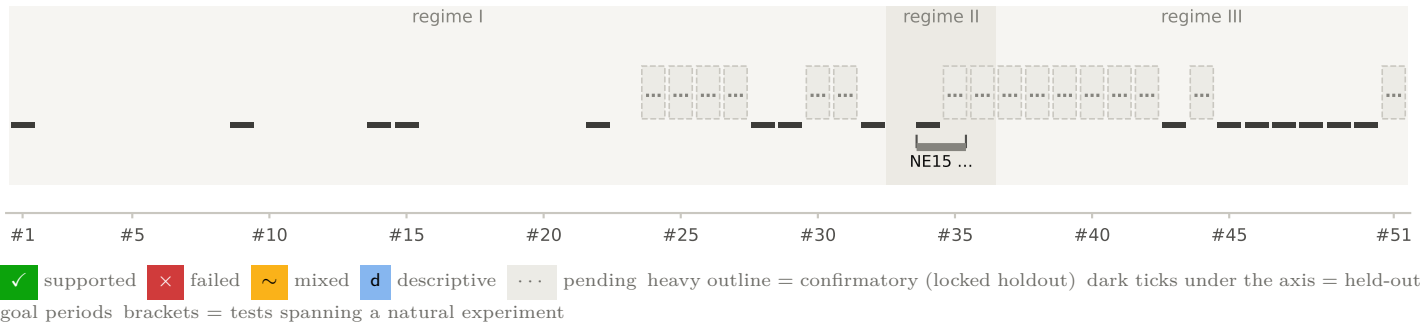
Not yet summarized. The question: When an agent takes its next turn, it faces k unread messages. Does the probability of responding to a given one fall as ~1/k (a fixed attention budget), making the mean-field J/N normalization literal? Does this micro-mechanism explain H03's finding that per-pair triggering and the branching ratio fall with N, and H05's rise in within-room coupling after the split (HH90)? Practical payoff: how coupling scales with swarm size and channel design.

PHYSICAL MODEL

From: physics-models/01-inverse-ising (mean-field J/N), 09-hawkes (cross-excitation per pair), 02-nonequilibrium-ising. Rival models: constant per-message response (no budget), saturating response, and recency-only (only the latest message matters). Uptake model (H18 variant, written 2026-10-03). At talk turn τ of recipient i on day d, each pending message m is "taken up" independently with hazard $\lambda_m = \theta_{\{i,d\}} \cdot h_m$

Pending.

GOAL PERIODS TESTED 17 pending



OBSERVABLES

Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

WHAT THE FIGURE SHOWS

Pending.

H19 One curve for all periods: per-period loop gains collapse onto a function of an operational control parameter

not yet rated

level: hypothesis · pipeline checks 86% · updated —

running. Exploratory round 1 done (2026-10-04): the pre-registered collapse P1 FAILED, with no common curve: equal-time activity gains rise with x_{att} while Hawkes talk gains fall, and the regime-only rival wins out of sample. Within the talk channel, the H...

WHAT HAPPENED

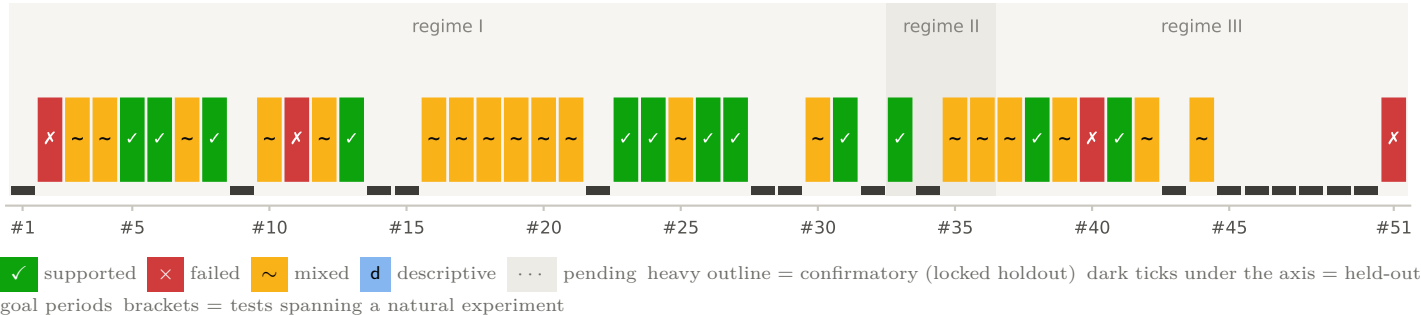
Not yet summarized. The question: Every hypothesis now produces per-goal-period coupling estimates: H02 Curie–Weiss βJ_0 , H03 branching ratio n and fast cross-triggering, H04 loop gain K , H05 J_{in}/J_{out} , and soon H10–H18. Do they collapse onto a single function of an operational control parameter, such as N , messages per agent-turn, hours per day, rooms, or the coupling mode of the goal? A collapse would be the swarm’s phase diagram and would predict coupling from operating conditions. Practical payoff: forecasting a swarm’s collective regime from its configuration.

PHYSICAL MODEL

From: physics-models/01-inverse-ising and 09-hawkes (mean-field loop gain $\beta J_0(1-m^2)$, branching ratio n), 02-nonequilibrium-ising. Finite-size scaling and data-collapse methodology. Hierarchical (partial-pooling) meta-regression across periods is the allowed exception (d) of the per-period rule: every period keeps its own estimate, and the meta-regression only relates those estimates to the period’s control parameters

Pending.

GOAL PERIODS TESTED 12 supported, 4 failed, 19 mixed



OBSERVABLES

Figure pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

WHAT THE FIGURE SHOWS

Pending.

H20 Aging: day-to-day content autocorrelation depends on time since kickoff, not just the lag

not yet rated

level: hypothesis · pipeline checks 57% · updated —

running (exploratory round 1). Observables, null and predictions written 2026-10-03, before any real-data run (below).

WHAT HAPPENED

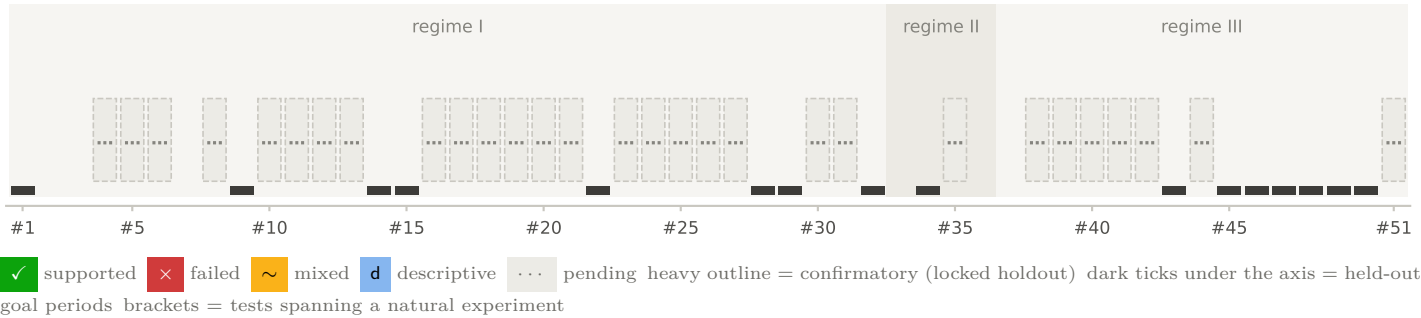
Not yet summarized. The question: Agents restart every day with memory, which is like field cooling. Does the similarity of what an agent (or the swarm) says on day t_w and day $t_w + \tau$ depend on the waiting time t_w since the goal's kickoff, as in spin-glass aging ($C(t_w + \tau, t_w)$ decays more slowly for larger t_w)? Or only on τ (stationary)? Do longer goals age more? Practical payoff: how fast a long-running swarm 'forgets' or locks in, and whether old swarms become rigid.

PHYSICAL MODEL

From: physics-models/01-inverse-ising (SK / glassy dynamics, aging and two-time correlation functions; fluctuation-dissipation violations), 11-vector-spins (content as vector spins; center and whiten). Rivals: stationary exponential relaxation; simple drift toward a fixed goal direction (field-driven relaxation with no aging). Hypothesis-specific variant. Each agent's day state is a vector spin in the regime's whiten

Pending.

GOAL PERIODS TESTED 29 pending



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

H21 The debate week (#12) is a two-sublattice antiferromagnet

not yet rated

level: hypothesis · pipeline checks 43% · updated —

running (exploratory round 1, 2026-10-03). Promoted 2026-10-03 by Vivian from HH103.

WHAT HAPPENED

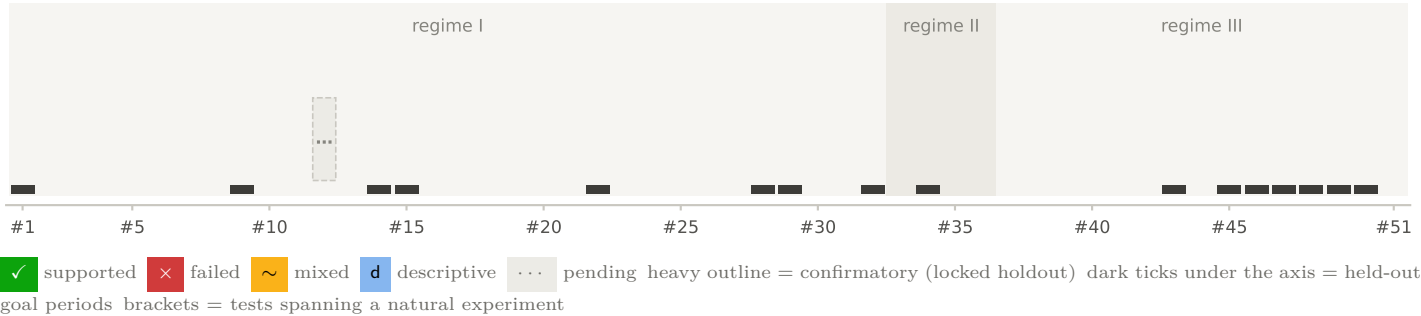
Not yet summarized. The question: In #12 the agents formed two teams to debate, with one agent judging. In content space, do the teams order as two sublattices: aligned within a team, anti-aligned across teams along the debate axis? Is the staggered magnetization $m_A - m_B$ the order parameter? Does the judge act as an external field that is zero during the debate and switched on at the verdict, collapsing or flipping the order? Practical payoff: detecting factions or teams from content alone, which links to finding hidden teams (D8.3). What the record says about the structure (derived before any outcome; see [G12](G12/README.md)). #12 is not one long two-team match. It is a tournament of 10 Asian-Parliamentary debates (09-01

PHYSICAL MODEL

From: physics-models/11-vector-spins (vector spins on two sublattices), physics-models/10-potts (two-sublattice ordering; $q = 2$ antiferromagnetic Potts = graph 2-colouring), physics-models/01-inverse-ising (antiferromagnetic mean field: staggered susceptibility). The magnet, precisely. For debate d , each debater i carries a unit spin $s_i \in S^{\{n-1\}}$ ($n = 32$ whitened embedding dimensions; definition below). Sublattice A

Pending.

GOAL PERIODS TESTED 1 pending



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

H22 Private, conflicting goals make #51 a spin glass

not yet rated level: hypothesis · pipeline checks 57% · updated —

exploratory round 1. Observables, nulls and predictions written 2026-10-04 00:15 UTC (2026-10-03 PT), before any real-data run (below).

WHAT HAPPENED

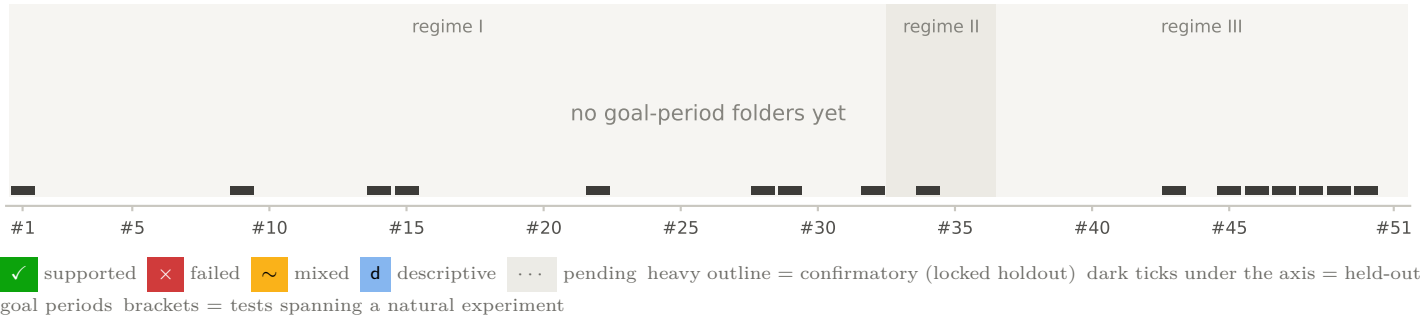
Not yet summarized. The question: In #51 each agent maximizes a privately assigned goal, and some goals conflict. Do the couplings between agents carry random signs, with frustration (unsatisfied triangles) well above shared-objective weeks? Couplings here means content alignment beyond the field, and talk coupling. Are the dynamics glassy: many metastable configurations, and overlaps between days that decay slowly and depend on history? Aging belongs to H20. Practical payoff: how mixed-motive swarms behave.

PHYSICAL MODEL

From: physics-models/01-inverse-ising (Sherrington–Kirkpatrick / spin-glass phenomenology: frustration, Edwards–Anderson overlap, P(q), SK phase diagram via moment matching, HH87), 11-vector-spins (content as O(n) spins), 10-potts. Block / mean-field statistics where pairwise inference is weak (H02); H05’s block_J reused by import. ### The magnet dictionary (H22 variant) | Magnet | Village (#51) | Dataset fields |

Pending.

GOAL PERIODS TESTED none yet



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.

H23 Distilling the village into a fine-tuned leader copies its vocabulary but transforms its plans

not yet rated

level: hypothesis · pipeline checks 43% · updated —

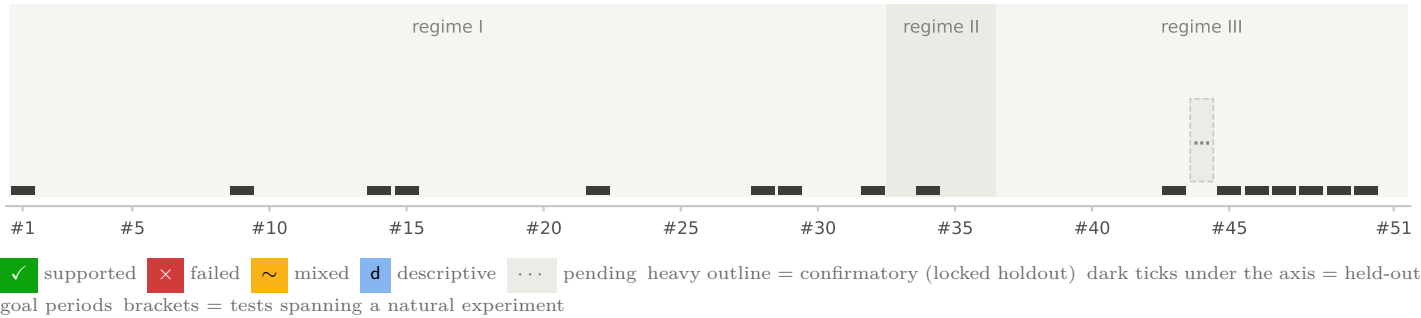
exploratory round 1 running (2026-10-03). Pipeline reconstructed; observables, null and prediction written below before any statistic was computed on real data.

WHAT HAPPENED
Not yet summarized. The question: In #44 the #best agents built training data and fine-tuned a leader model. A temporary fine-tuned leader (roster agent 28) ran on 05-28 and 05-29 inside #44; the final leader (agent 30) ran in #45 ((held out)). Using Kolchinsky-style copy vs transformation information: what did distillation copy from the corpus (vocabulary, phrasing, topics) and what did it transform (plans, priorities)? Is the leader’s content closer to the training corpus than to its base model’s field? Practical payoff: what survives when a swarm is distilled into one model.

PHYSICAL MODEL
From: physics-models/08-copying-vs-transformation ($I(X;Y) = I_{\text{copy}} + I_{\text{transform}}$), physics-models/04-semantic-information, 11-vector-spins (embedding-level comparisons). H23 variant: copy information (distillation variant). A channel from the corpus (X) to the leader’s live output (Y) over a shared alphabet, at three levels of coarse-graining: - phrase level: the alphabet is word n-grams (agent names → <agent>, numbe

Pending.

GOAL PERIODS TESTED 1 pending



OBSERVABLES
Figure pending.

WHAT THE FIGURE SHOWS
Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS
Pending.
Bottom line. *Pending.*

H24 In the forecast week (#21), switching from independent drafts to comparison switches the coupling on

not yet rated

level: hypothesis · pipeline checks 43% · updated —

running (exploratory round 1). Switch-on time determined from the record and predictions written 2026-10-03, before any real-data alignment, dispersion or coupling statistic was computed (below).

WHAT HAPPENED

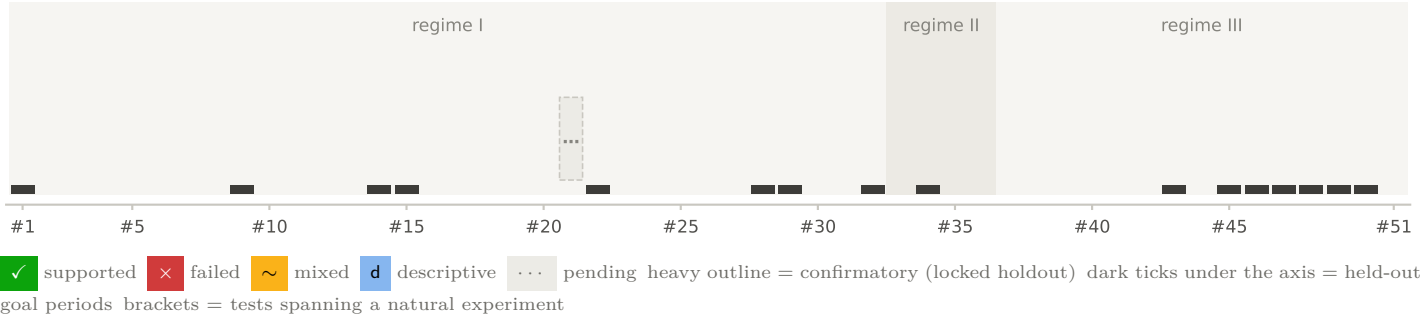
Not yet summarized. The question: In #21 the agents forecast AI abilities and effects. If they first drafted independently and then compared, the coupling was effectively switched on mid-week. Do their forecasts (as vectors in content space, and as numeric forecasts where extractable) show low alignment during drafting and a jump in alignment, with shrinking variance (herding), once comparison starts? Is the crowd's diversity lost faster than independent updating would predict? Practical payoff: forecast aggregation in swarms, wisdom of crowds vs herding.

PHYSICAL MODEL

From: physics-models/11-vector-spins ($O(n)$ and soft-spin Gaussian versions: alignment jump at coupling switch-on; mean-field βJ_0 before vs after), 01-inverse-ising (Curie–Weiss analogue for numeric forecasts as soft spins).
- Content ($O(n)$, $n = 32$): each agent's statements are noisy draws around a unit state $s_i(t)$. Before switch-on, $s_i = \text{unit}(h_g + h_i + \text{noise})$ (common field plus agent fields, no coupling). After s

Pending.

GOAL PERIODS TESTED 1 pending



OBSERVABLES

Figure pending.

WHAT THE FIGURE SHOWS

Pending.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Bottom line. Pending.